Linear Programming Formulation

Optimization / Program : $\max_x / \min_x \; f(x)$ s.t. $g(x) \geq 0$ → Linear functions in LP

LP aims to find the optimal value function or an optimal policy by directly solving an optimization problem.

Bellman optimality equation:

$$V^*(s) = \max_{a \in A} \left[R(s,a) + \gamma \, \mathbb{E}_{s' \sim P(\cdot|s,a)} [V^*(s')] \right] \quad \forall s \in S$$

$x_1, x_2, \ldots, x_n$ and $x^* = \max_i x_i$ $\xrightarrow{\text{as an LP}}$ $x^* = \min x$ s.t. $x \geq x_i \;\; \forall i \in \{1, 2, \ldots, n\}$

Note: Based on the Bellman optimality equation,

$$V^*(s) \geq R(s,a) + \gamma \, \mathbb{E}_{s' \sim P(\cdot|s,a)} [V^*(s')] \quad \forall s \in S, \forall a \in A;$$

because in each state, for the optimal action(s), $=$ holds and for the suboptimal action(s), $>$ holds.

Primal LP

Primal LP finds V^* by $\underbrace{\text{searching for } V \in \mathbb{R}^{|S|}}_{\text{in the space of value functions}}$ that satisfies the Bellman optimality equation.

$$\min_{V \in \mathbb{R}^{|S|}} \sum_{s \in S} \mu_0(s) V(s)$$

→ $\mu_0 \in \Delta(S)$ s.t. $\mu_0(s) > 0 \;\; \forall s \in S$

$$\text{subject to} \quad V(s) \geq R(s,a) + \gamma \, \mathbb{E}_{s' \sim P(\cdot|s,a)} [V(s')] \quad \forall s \in S, \forall a \in A$$

Once we find v^* by solving the primal LP $\longrightarrow$

we can find an optimal policy

$$\pi^*(s) = \underset{a \in A}{\arg\max} \underbrace{\left[R(s,a) + \gamma \, \mathbb{E}_{s' \sim P(\cdot|s,a)} [v^*(s')] \right]}_{Q^*(s,a)} \quad \forall s \in S$$

* The primal LP has $\begin{cases} |S| \text{ unknown variables} \\ |S||A| \text{ constraints} \end{cases}$

Note: Any feasible solution of the primal LP will be an upper bound on v^* (based on the previous "note"). Therefore, minimizing the objective function, minimizes this upper bound so that it matches v^* (making the inequality constraints $\underbrace{\text{active}}_{=}$ for the optimal actions at all states).

Duality for LPs: For an LP, one can write a dual LP.

Primal LP		Dual LP
$\min\limits_{x} \; c^T x$ subject to $Ax \geq b$	$\longrightarrow$	$\max\limits_{y} \; b^T y$ subject to $A^T y = c$ $y \geq 0$

- $x \in \mathbb{R}^n$ unknown variables in the primal LP
- $y \in \mathbb{R}^m$ unknown variables in the dual LP
- $c \in \mathbb{R}^n$
- $A \in \mathbb{R}^{m \times n}$
- $b \in \mathbb{R}^m$

strong duality: If the primal LP has a (finite) solution x^*, the dual LP also has a (finite) solution y^* such that $c^T x^* = b^T y^*$, i.e.,

$$\begin{array}{ll} \min\limits_{x} & c^T x \\ \text{subject to} & Ax \geq b \end{array} \quad = \quad \begin{array}{ll} \max\limits_{y} & b^T y \\ \text{subject to} & A^T y = c \\ & y \geq 0 \end{array} .$$

Dual LP

in the space of policies/occupancy measures

Dual LP finds π^* by searching for $\nu \in \mathbb{R}^{|S||A|}$, occupancy measure corresponding to a policy, that maximizes the value function.

$$= \sum_{s \in S} \mu_0(s)\, V(s)$$

$$\boxed{\begin{aligned} &\max_{\nu \geq 0} \quad \sum_{\substack{s \in S \\ a \in A}} \nu(s,a)\, R(s,a) \\ &\text{subject to} \quad \sum_{a \in A} \nu(s,a) = \mu_0(s) + \gamma \sum_{\substack{s' \in S \\ a' \in A}} P(s|s',a')\, \nu(s',a') \quad \forall s \in S \end{aligned}}$$

Once we find ν^* by solving the dual LP $\longrightarrow$ we can find an optimal policy

$$\pi^*(a|s) = \begin{cases} \dfrac{\nu(s,a)}{\sum_{a' \in A} \nu(s,a')} & \text{if } \sum_{a' \in A} \nu(s,a') \neq 0 \\ \text{arbitrary policy} & \text{otherwise} \end{cases} \qquad \forall s \in S$$

* The dual LP has $\begin{cases} |S||A| & \text{unknown variables} \\ |S| & \text{constraints} \end{cases}$

Note: Any feasible solution of the dual LP will be a proper occupancy measure, i.e., $\exists \pi$ s.t. $\nu^\pi = \nu$.
The objective maximizes the value of such a policy.

potentially stochastic

Occupancy Measures: For a policy π,

the state occupancy measure $\rho^{\pi}_{\mu_0} : \mathcal{S} \to \mathbb{R}$, and

the state-action occupancy measure $\nu^{\pi}_{\mu_0} : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$

are defined as the cumulative discounted visitation probability of states and state-action pairs, respectively. Formally,

$\mathbb{E}_{\mu_0, \pi, P}\left[\sum_{t=0}^{\infty} \gamma^t \mathbb{1}[S_t = s]\right]$

$$\rho^{\pi}_{\mu_0}(s) = \sum_{t=0}^{\infty} \gamma^t \mathbb{P}(S_t = s \mid \mu_0, \pi, P),$$

$$\nu^{\pi}_{\mu_0}(s,a) = \sum_{t=0}^{\infty} \gamma^t \mathbb{P}(S_t = s, A_t = a \mid \mu_0, \pi, P).$$

$\mathbb{E}_{\mu_0, \pi, P}\left[\sum_{t=0}^{\infty} \gamma^t \mathbb{1}[S_t = s, A_t = a]\right]$

The normalized occupancy measure are defined as

$$\bar{\rho}^{\pi}_{\mu_0}(s) = (1-\gamma)\, \rho^{\pi}_{\mu_0}(s),$$

$$\bar{\nu}^{\pi}_{\mu_0}(s,a) = (1-\gamma)\, \nu^{\pi}_{\mu_0}(s,a).$$

Note: Informally, the state occupancy measure and the state-action occupancy measure characterize how often (in a discounted way) a state and a state-action pair are visited following a policy.

Notes about LP formulation:

- LP formulation leads to elegant theoretical analysis.
- Depending on the solver, dual LP may be more efficient to solve.
- Solutions to LP include → simplex method
  → interior-point method
- Using a polynomial solver for these LP formulations ensures a polynomial solution.
- Using an interior-point method for solving the LP has been proven to be strongly polynomial.
- PI is equivalent to applying a (block) simplex algorithm over LP.
- The computational complexity of the LP solution is $O\left(|S|^4 |A|^4 \log \frac{|S|}{1-\gamma}\right)$. (It has been improved.)